



APRIL/ 2025

REGIONAL MARITIME UNIVERSITY
MARINE ENGINEERING
BSc MARINE ENG. (BME 28 B)/ BSc MECHANICAL ENG. (BMT 28 B)
END-OF-SEMESTER EXAMINATIONS
BASIC MECHANICS

INSTRUCTIONS: Attempt any five (5) questions. *All questions carry equal marks.*
TIME ALLOWED: 3 hours

QUESTION 1 [20 marks]

The coefficients of static and kinetic friction between the 100-kg block and the inclined plane, as shown in Figure Q1, are 0.30 and 0.20, respectively. Determine:

- the friction force F acting on the block when P is applied with a magnitude of 200 N to the block at rest,
- the force P required to initiate motion up the incline from rest, and
- the friction force F acting on the block if $P = 600$ N. [$g = 9.81 \text{ m/s}^2$]

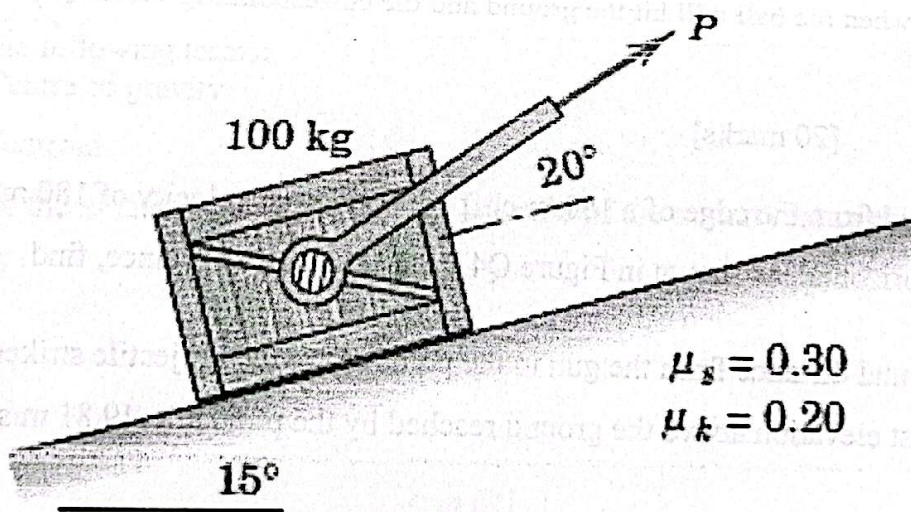


Figure Q1

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QUESTION 2 [20 marks]

The position coordinate of a particle which is confined to move along a straight line is given by $s = 2t^3 - 24t + 6$, where s is measured in meters from a convenient origin and t is in seconds. Determine:

- the time required for the particle to reach a velocity of 72 m/s from its initial condition at $t = 0$ s,
- the acceleration and displacement of the particle when the velocity, $v = 30$ m/s, and
- the net displacement of the particle during the interval from $t = 1$ s to $t = 4$ s.

Handwritten notes for Question 2:

$$v = 72$$

$$s = 2t^3 - 24t + 6$$

$$s = 0$$

$$v = u + at$$

$$v = 72$$

$$6t = 72$$

QUESTION 3 [20 marks]

A ball is tossed with a velocity of 10 m/s directed vertically upward from a window located 30 m above the ground. Knowing that the acceleration of the ball is constant and equal to 9.81 m/s^2 downward and neglecting air resistance, determine

- the highest elevation reached by the ball and the corresponding time t , to reach that elevation,
- the time when the ball will hit the ground and the corresponding velocity.

Handwritten notes for Question 3:

$$a = \frac{dv}{dt}$$

$$30 = 0 + at$$

$$30 = 0 + 9.81t$$

QUESTION 4 [20 marks]

A projectile is fired from the edge of a 150-m cliff with an initial velocity of 180 m/s at an angle of 30° with the horizontal, as shown in Figure Q4. Neglecting air resistance, find

- the horizontal distance from the gun to the point where the projectile strikes the ground,
- the greatest elevation above the ground reached by the projectile. [9.81 m/s^2]

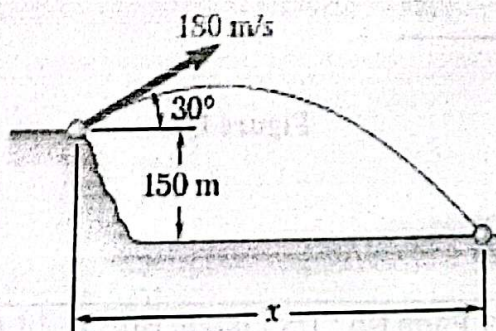


Figure Q4

Handwritten notes for Question 4:

$$H = \frac{2u^2 \sin^2 \theta}{g}$$

$$v^2 = u^2 + 2as$$

$$v = u + at$$

$$0 = 180 - 9.81t$$

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QUESTION 5 [20 marks]

Determine the resultant **R** of the three tension forces acting on the eye bolt shown in Figure Q5. Find the magnitude of **R** and the angle which **R** makes with the positive x-axis, measured counter-clockwise.

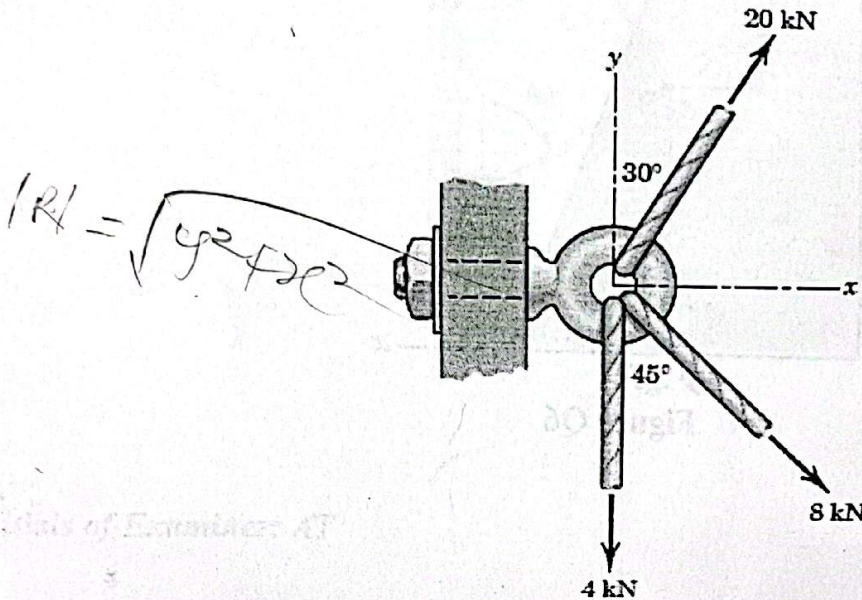


Figure Q5

QUESTION 6 [20 marks]

- Define the following terms:
 - Centre of gravity
 - Centroid
- Determine the x- and y-coordinates of the centroid of the trapezoidal area shown in Figure Q6.

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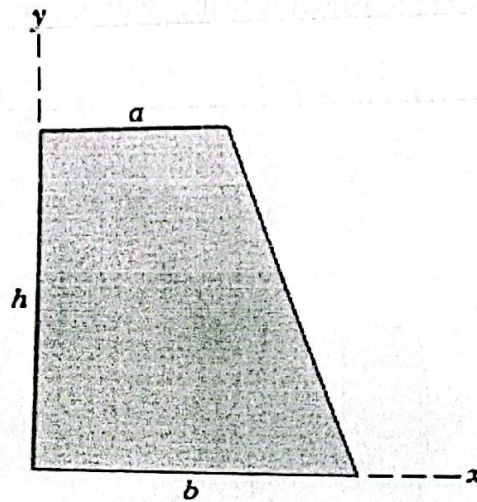


Figure Q6

QUESTION 7 [20 marks]

Determine the equivalent concentrated load(s) and external reactions for the simply supported beam which is subjected to the distributed load shown in Figure Q7.

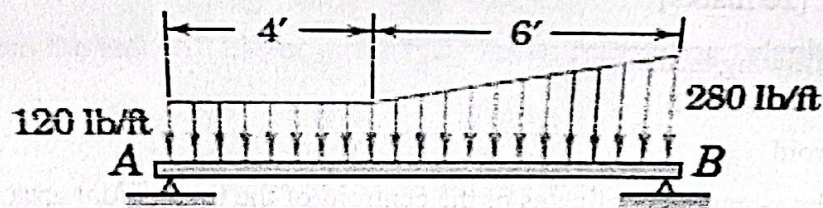


Figure Q7

QUESTION 8 [20 marks]

Two cables tied together at C are loaded as shown in Figure Q8. Knowing that $Q = 60$ lb, determine the tension in cable

- AC, and
- BC.

$$(b) \quad L = \frac{b^2 - ab - a^2}{a^2 + b^2}$$

$$\frac{1}{6} h (a^2 + b^2 - 2ab)$$

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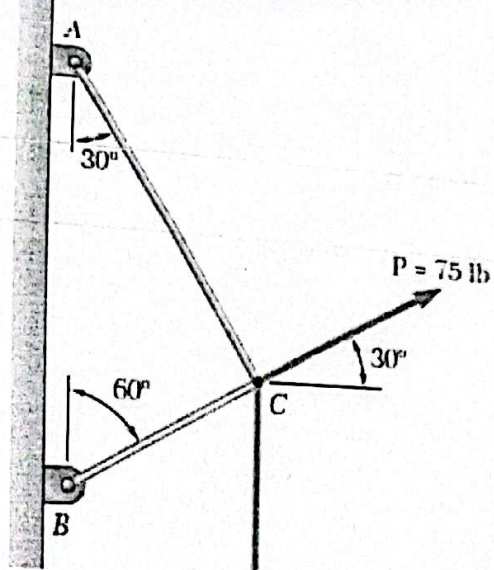
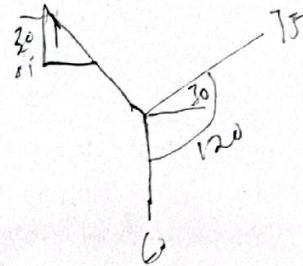


Fig. Q8

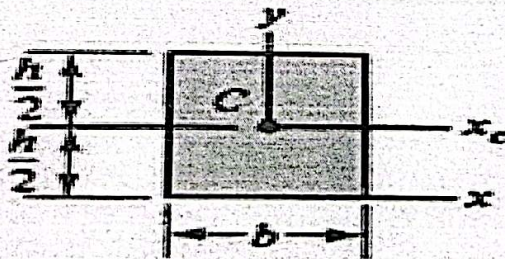


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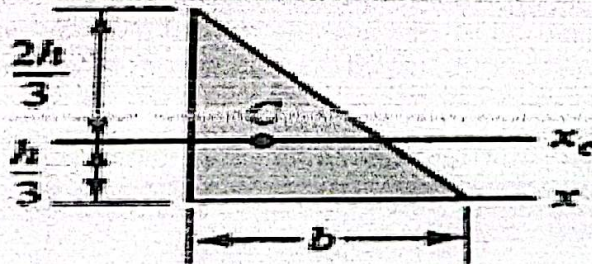
Property Table

Shapes

Rectangle



Triangle



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